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MATH 246A

Homework 2
Due: TBD

Problem 3.1

Let $D = \overline{B}(a, r)$ be a closed Euclidean disk in $\mathbb{C}$ of radius $r > 0$ centered at $a \in \mathbb{C}$.
(a) Suppose $z_1, ..., z_n$ are points in D, and $\lambda_1, ..., \lambda_n$ are real numbers in $[0, 1]$ with $\lambda_1 + ... + \lambda_n = 1$. Show that $\lambda_1 z_1 + ... + \lambda_n z_n \in D$.
(b) Suppose $P(z) = \prod_{k=1}^{n}(z - z_k)$ is a complex polynomial whose zeros $z_1, ..., z_n$ lie in D. Show that the zeros of P' also lie in D.

Solution. **(a)**: Suppose $z_1, ..., z_n \in D$ and $\lambda_1, ..., \lambda_n \in [0, 1]$ where

$$\lambda_1 + ... + \lambda_n = 1.$$

Then the claim follows from a immediate application of the triangle inequality:

$$\begin{aligned}\left|\sum_{k=1}^{n} \lambda_k z_k\right| &= \sum_{i=1}^{n} \lambda_k |z_k| \\ &\leq r \sum_{k=1}^{n} \lambda_k \\ &= r\end{aligned}$$

(b):

Problem 3.2

(a) Let $S^1 := \{w \in \mathbb{C} : |w| = 1\}$. Show that $f : \mathbb{R} \to S^1$, $t \mapsto e^{it}$, is a covering map of $\mathbb{R}$ onto S^1, i.e., for each $w_0 \in S^1$ there is an open neighborhood $U \subseteq S^1$ of w_0 such that $f^{-1}(U) = \bigcup_{k\in\mathbb{Z}} U_k$ for pairwise disjoint open sets $U_k \subseteq \mathbb{R}$ with $f|_{U_k}$ a homeomorphism of U_k onto U for each $k \in \mathbb{Z}$.
(b) Show that $\exp : \mathbb{C} \to \mathbb{C}^* = \mathbb{C} \setminus \{0\}$ is a covering map.

Solution. Throughout, Arg denotes the principal branch of the argument, $\text{Arg}(z) \in (-\pi, \pi]$.

(a): $\forall w \in S^1$, $w = e^{it_k}$ where

$$t_k := \theta + 2\pi k, \quad \text{where } \theta := \text{Arg}(w) \in (-\pi, \pi], k \in \mathbb{Z}.$$

Take $U = B\left(w, \sqrt{2}\right) \cap S^1$. Take $U_k := \left(\theta + \frac{\pi(4k-1)}{2}, \theta + \frac{\pi(4k+1)}{2}\right) = (\theta + 2\pi k - \frac{\pi}{2}, \theta + 2\pi k + \frac{\pi}{2})$. Note that $U_k \subset (\theta + \pi(2k-1), \theta + \pi(2k+1))$. Note that for distinct $a, b \in \mathbb{Z}$, $(\theta + \pi(2a - 1), \theta + \pi(2a + 1)) \cap (\theta + \pi(2b - 1), \theta + \pi(2b + 1)) = \emptyset$. Note that $f(U_k) = U$. Suppose that $\alpha, \beta \in U_k$ where $f(\alpha) = e^{i\alpha} = e^{i\beta} = f(\beta) \Rightarrow |\alpha - \beta| = 2\pi a$ where $a \in \mathbb{N}$. But the length of U_k, is π hence $a = 0$ and $\alpha = \beta$. Thus $f|_{U_k}$ is injective onto U hence $f|_{U_k}$ is a continuous bijection. It remains to show the inverse is continuous. Let $H := \{z \in \mathbb{C} : \text{Re}(z) > 0\}$. On H we have $\text{Arg}(z) \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $\sin(\text{Arg}(z)) = \text{Im}\,\frac{z}{|z|}$. Since $\arcsin : [-1, 1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$ inverts sin there,

$$\operatorname{Arg}(z) = \arcsin\left(\operatorname{Im}\frac{z}{|z|}\right) \quad (z \in H),$$

which is a composition of continuous functions, hence Arg is continuous on H. Define

$$g_k : U \to \mathbb{R}, \quad g_k(w) := \theta + 2\pi k + \operatorname{Arg}\left(we^{-i\theta}\right).$$

If $w = e^{is} \in U$ with $|s - \theta| < \frac{\pi}{2}$, then $we^{-i\theta} = e^{i(s-\theta)} \in H$ with $s - \theta \in (-\frac{\pi}{2}, \frac{\pi}{2}) \subseteq (-\pi, \pi]$, so $\operatorname{Arg}\left(we^{-i\theta}\right) = s - \theta$ and $g_k(w) = s + 2\pi k \in U_k$. Hence g_k is well defined, continuous, and $g_k(U) \subseteq U_k$. (We apply Arg to $we^{-i\theta} \in H$ rather than to w because Arg is discontinuous at -1, and -1 may lie in U.) Moreover

$$f(g_k(w)) = e^{i\theta}e^{2\pi ik}e^{i\operatorname{Arg}\left(we^{-i\theta}\right)} = e^{i\theta} \cdot we^{-i\theta} = w,$$

i.e. $f|_{U_k} \circ g_k = \mathrm{id}_U$. Since $f|_{U_k} : U_k \to U$ is a bijection, $g_k = \left(f|_{U_k}\right)^{-1}$. Thus $f|_{U_k}$ is a continuous bijection with continuous inverse, i.e. a homeomorphism of U_k onto U.

(a): exp is surjective this is just a matter of checking using polar form. $\forall w \in \mathbb{C}^\times$, $w = re^{iy_k}$ where

$$y_k := \theta + 2\pi k, \theta \in (-\pi, \pi)$$

and $r > 0$. Let $x = \log(r)$. Let $U = \left\{z \in \mathbb{C}^\times : \frac{z}{|z|} \in B\left(\frac{w}{|w|}, \sqrt{2}\right)\right\}$

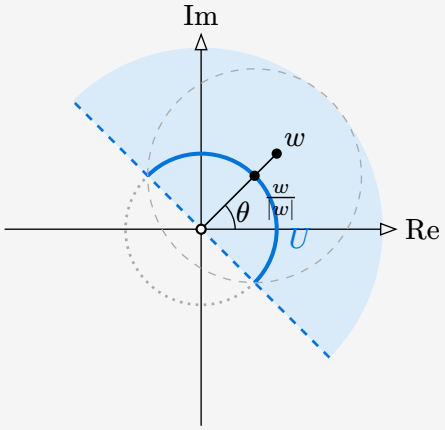


Figure 1: Let $w = 1 + i$. Then U is shown.

We have the following inverse image for U.

$$f^{-1}(U) := \bigsqcup_{k \in \mathbb{Z}} \mathbb{R} \times \left(\theta + 2\pi k - \frac{\pi}{2}, \theta + 2\pi k + \frac{\pi}{2}\right).$$

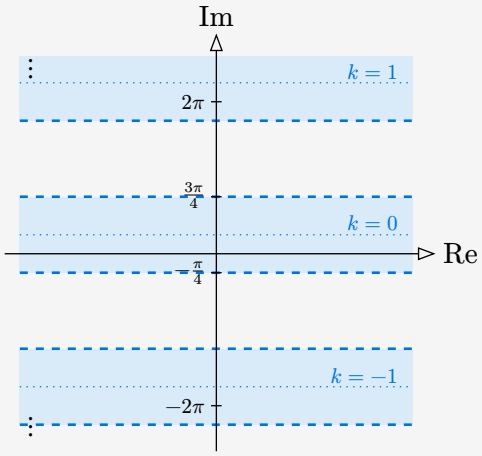


Figure 2: $\exp^{-1}(U)$ for $w = 1 + i$ ($\theta = \frac{\pi}{4}$): the disjoint open horizontal strips $\mathrm{Im}(z) \in (\theta + 2\pi k - \frac{\pi}{2}, \theta + 2\pi k + \frac{\pi}{2})$, each of height π, shown for $k = -1, 0, 1$. The dotted centre line goes onto the ray through w.

Much like section 1, f is injective and surjective. The inverse $U \to \mathbb{R} \times U_k$ is given by $w \to \log(|w|) + ig_{k(z)}$ everything is continuous and behaves nicely

Problem 3.3

Let $P(z) = a_0 + a_1 z + ...$ and $Q(z) = b_0 + b_1 z + ...$ be polynomials with complex coefficients. Show that if $P(z) = Q(z)$ for all z, then $a_0 = b_0$, $a_1 = b_1$, etc.

Solution. $P(x) - Q(x) = 0$ everywhere. Then we see that $(P - Q)' = 0 \Rightarrow P - Q$ is a constant and hence 0. Thus

$$(P - Q)(x) = \sum_{k=0}^{n} (a_k - b_k) x^k = 0.$$

We know that $(P - Q)(0) = 0 \Rightarrow a_0 - b_0 = 0$. Then keep taking successive derivatives hence $P^{(k)}(0) = k!(a_k - b_k) = 0 \Rightarrow a_k = b_k$.

Problem 3.4

Let $U \subseteq \mathbb{C}$ be open and $z_0 \in U$. Suppose $f : U \to \mathbb{C}$ is differentiable at z_0 and $f(z_0) \neq 0$. Prove directly from the definitions that there is a neighborhood V of z_0 such that $g(z) = \frac{1}{f(z)}$ is defined on V, that g is differentiable at z_0, and that

$$g'(z_0) = -\frac{f'(z_0)}{f(z_0)^2}.$$

Hint: do not apply the quotient rule; establish this special case directly.

Solution. Note that g is continuous at z_0. $\forall \varepsilon > 0$, let $\delta_1 > 0$ such that

$$|z - z_0| < \delta_1 \Rightarrow \left| \frac{1}{f(z_0)} - \frac{1}{f(z)} \right| < \frac{\varepsilon |f(z_0)|}{2(|f'(z_0)| + 1)}.$$

Let $\delta_2 > 0$ such that

$$|z - z_0| < \delta_2 \Rightarrow |f(z) - f(z_0)| < \frac{|f(z_0)|}{2} \Rightarrow |f(z)| \geq \frac{|f(z_0)|}{2}$$

Let $\delta_3 > 0$ such that

$$|z - z_0| < \delta_3 \Rightarrow \left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \frac{\varepsilon |f(z_0)|^2}{4}.$$

Let $\delta = \min(\delta_1, \delta_2, \delta_3)$. Thus if $|z - z_0| < \delta$ then all three bounds are correct. Thus

$$\begin{aligned}
\left| \frac{g(z) - g(z_0)}{z - z_0} + \frac{f'(z_0)}{(f(z_0))^2} \right| &= \left| \frac{f(z_0) - f(z)}{f(z) f(z_0)(z - z_0)} + \frac{f'(z_0)}{(f(z_0))^2} \right| \\
&= \frac{1}{|f(z_0)|} \left| \frac{f'(z_0)}{f(z_0)} - \frac{f(z) - f(z_0)}{f(z)(z - z_0)} \right| \\
&\leq \frac{|f'(z_0)|}{|f(z_0)|} \left| \frac{1}{f(z_0)} - \frac{1}{f(z)} \right| + \frac{1}{|f(z_0)||f(z)|} \left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| \\
&< \frac{|f'(z_0)|}{|f(z_0)|} \cdot \frac{\varepsilon |f(z_0)|}{2(|f'(z_0)| + 1)} + \frac{2}{|f(z_0)|^2} \cdot \frac{\varepsilon |f(z_0)|^2}{4} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,
\end{aligned}$$

where in the third line we split $\frac{f'(z_0)}{f(z_0)} - \frac{f(z) - f(z_0)}{f(z)(z - z_0)} = f'(z_0)\left(\frac{1}{f(z_0)} - \frac{1}{f(z)}\right) + \frac{1}{f(z)}\left(f'(z_0) - \frac{f(z) - f(z_0)}{z - z_0}\right)$, and in the fourth we used $|f(z)| \geq \frac{|f(z_0)|}{2}$ from δ_2. Since $\varepsilon > 0$ was arbitrary, g is differentiable at z_0 with $g'(z_0) = -\frac{f'(z_0)}{(f(z_0))^2}$. Finally, δ_2 does not depend on ε, so $V \coloneqq B(z_0, \delta_2) \cap U$ is a neighborhood of z_0 on which $|f| \geq \frac{|f(z_0)|}{2} > 0$, hence $g = \frac{1}{f}$ is defined on V.

Problem 3.5

Let $U \subseteq \mathbb{C}$ be a region and $u : U \to \mathbb{R}$ harmonic. Suppose v and w are harmonic conjugates of u on U. Show that $v - w$ is constant on U.

Solution. Note that $f = u + iv$ and $g = u + iw$ are holomorphic. Thus f and g satisfy the Cauchy Riemann equations:

$$\begin{aligned}
u_x &= v_y = w_y \\
u_y &= -v_x = -w_x
\end{aligned}$$

Thus $v_y = w_y$ and $v_x = w_x$. Let $h = v - w$. We know that

$$\operatorname{grad}(h) = 0.$$

We want to show that h is a constant. We will show h is locally constant and hence is constant by path connectedness of U. Take $\vec{x} \in U$ (viewing as a element of $\mathbb{R}^2$). By openness of u, there exists $\delta > 0$ such that $B(x, \delta) \subseteq U$. $B(x, \delta)$ is convex and hence $\forall y \in B(x, \delta)$, there exists a line from y to x. By the mean value theorem, $\exists \theta \in [0, 1]$ such that

$$|h(x) - h(y)| \leq \text{grad}(h)(x + \theta(y - x)) \cdot (x - y) = 0.$$

Thus $h(x) - h(y) = 0 \Rightarrow h(x) = h(y)$. Thus h is locally constant. Now since h is locally constant and U is path connected, it must be constant. Thus $v - w$ is a constant function.

Problem 3.6

Let $U \subseteq \mathbb{C}$ be a region and $f \in \mathcal{H}(U)$. Using the Cauchy–Riemann equations, show:
(a) if $\text{Re}(f)$ or $\text{Im}(f)$ is constant on U, then f is constant.
(b) if $|f|$ is constant on U, then f is constant.

Solution. $f = u + iv$ holomorphic and where u and v are real valued.

1. It suffices to prove the statement for u constant. If v was constant, then consider $if = -v + iu$, which is constant and hence f is constant. We know the following

$$0 = u_x = v_y$$
$$0 = u_y = -v_x.$$

   Hence $\text{grad}(v)$ is 0. Now use multivariable mean value theorem to show that v is locally constant and hence constant. Thus f must be constant.
2. Suppose $|f|$ is constant on U. That means

$$\begin{aligned} c = |u + iv| &= (u + iv)\overline{(u + iv)} \\ &= u^2 + v^2. \end{aligned}$$

Thus

$$0 = 2uu_x + 2vv_x$$
$$0 = 2uu_x - 2vu_y$$

and

$$0 = 2uu_y + 2vu_x.$$

Let

$$M(x, y) = \begin{pmatrix} u(x, y) & -v(x, y) \\ v(x, y) & u(x, y) \end{pmatrix}.$$

We have that

$$M \begin{pmatrix} u_x(x, y) \\ u_y(x, y) \end{pmatrix} = 0.$$

Note that $\det(M) = |f|^2$.
1. $|f|^2 = 0$, then $f = 0$ and we are done.
2. $|f|^2 \neq 0$: $M(x,y)$ is invertible and $\operatorname{grad}(u) = 0 \Rightarrow u$ locally constant, hence constant, and f constant.

Problem 3.7

Let $U, V \subseteq \mathbb{C}$ be open, $h : V \to \mathbb{R}$ harmonic, and $f : U \to V$ holomorphic. Show that $g := h \circ f : U \to \mathbb{R}$ is harmonic. *Hint:* you may disregard smoothness issues. To show $\Delta g = 0$, use Wirtinger derivatives throughout.

Solution. Note the following

$$4\partial_{\bar{z}}\partial_z = 4\partial_z\partial_{\bar{z}} = \big(\partial_x - i\partial_y\big)\big(\partial_x + i\partial_y\big) = \partial_x^2 + \partial_y^2 = \Delta.$$

Consider

$$\begin{aligned}\Delta(h \circ f) &= 4\partial_z\partial_{\bar{z}}(h \circ f)\\ &= 4\partial_z((\partial_{\bar{z}}(h) \circ f)\partial_{\bar{z}}(f))\\ &= 0 \quad (\text{Because } f \in \mathcal{H}(U))\end{aligned}$$

Thus $h \circ f$ is harmonic.

Problem 3.8

Let $U \subseteq \mathbb{C}$ be open. A C^2 function $s : U \to \mathbb{R}$ is *subharmonic* if $\Delta s \geq 0$ on U. Show that if $f : U \to \mathbb{C}$ is holomorphic, then $s(z) = \log(1 + |f(z)|^2)$ is subharmonic on U.

Solution. Note the following:

$$\begin{aligned}\partial_{\bar{z}}|f|^2 &= \partial_{\bar{z}}\big(f\overline{f}\big)\\ &= \partial_{\bar{z}}(f)\overline{f} + \partial_{\bar{z}}\big(\overline{f}\big)f\\ &= \overline{\partial_z(f)}f\end{aligned}$$

$$\partial_z|f|^2 = \partial_z f\overline{f}$$

$$\begin{aligned}\Delta|f|^2 &= 4\partial_z\partial_{\bar{z}}\big(f\overline{f}\big)\\ &= 4|\partial_z f|^2\end{aligned}$$

Also $\partial_z\partial_{\bar{z}}|f|^2 = |\partial_z f|^2$ and $\big|\partial_z|f|^2\big|^2 = |f|^2|\partial_z f|^2$. Since f is smooth and $1 + |f|^2 \geq 1$, s is C^2. Then

$$\begin{aligned}\Delta s = 4\partial_z\partial_{\bar{z}}(s) &= 4\partial_z\left(\frac{1}{1+|f|^2}\cdot\partial_{\bar{z}}(|f|^2)\right)\\ &= -4(1+|f|^2)^{-2}\partial_{\bar{z}}(|f|^2)\partial_z(|f|^2) + 4(1+|f|^2)^{-1}\partial_z\partial_{\bar{z}}|f|^2\\ &= -4(1+|f|^2)^{-2}|f|^2|\partial_z f|^2 + 4(1+|f|^2)^{-1}|\partial_z f|^2\\ &= \frac{4|\partial_z f|^2}{(1+|f|^2)^2}(1+|f|^2-|f|^2) = \frac{4|\partial_z f|^2}{(1+|f|^2)^2} \geq 0.\end{aligned}$$

Problem 3.9

Let u be a C^2 function on an open subset of $\mathbb{R}^2$, viewed as $u(x,y)$ and, via $x = r\cos\varphi$, $y = r\sin\varphi$, as a function of (r,φ). Subscripts denote partial derivatives.
(a) Show that $u_r = \cos(\varphi)u_x + \sin(\varphi)u_y$ and $u_\varphi = -r\sin(\varphi)u_x + r\cos(\varphi)u_y$.
(b) Find similar expressions for u_{rr} and $u_{\varphi\varphi}$.
(c) Using (a) and (b), show that $\Delta u = u_{xx} + u_{yy} = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\varphi\varphi}$.

Solution. **(a)**: By the multivariable chain rule we know that

$$\begin{aligned}u_r &= u_x x_r + u_y y_r\\ &= \cos(\varphi)u_x + \sin(\varphi)u_y\end{aligned}$$

and

$$\begin{aligned}u_\varphi &= u_x x_\varphi + u_y y_\varphi\\ &= -ru_x\cos(x) + r\sin(\varphi)u_y\end{aligned}$$

(b):

$$\begin{aligned}u_{rr} &= \cos(\varphi)\big(u_{xx}\cos(\varphi) + u_{xy}\sin(\varphi)\big) + \sin(\varphi)\big(u_{xy}\cos(\varphi) + u_{yy}\sin(\varphi)\big)\\ &= u_{xx}\cos^2(\varphi) + 2u_{xy}\cos(\varphi)\sin(\varphi) + u_{yy}\sin^2(\varphi)\end{aligned}$$

Problem 4.1

Let $\Omega \subseteq \mathbb{C}$ be convex and $f \in \mathcal{H}(\Omega)$. Show that if $\mathrm{Re}(f') > 0$ on Ω, then f is injective.

Solution. Fix $x \in \Omega$. Let $y \in \Omega$. Consider $\varphi(t) := x + (y-x)t$.

$$\begin{aligned}f(y) &= f(x) + \int_0^1 f'(\varphi(t))(y-x)\,\mathrm{d}t\\ &= f(x)\end{aligned}$$

Problem 4.2

A simple special case of Fubini's theorem.

(a) Let $[a,b],[c,d] \subseteq \mathbb{R}$ be compact intervals and $F : [a,b] \times [c,d] \to \mathbb{C}$ continuous. Show that $x \mapsto \int_c^d F(x,y)dy$ and $y \mapsto \int_a^b F(x,y)dx$ are continuous on $[a,b]$ and $[c,d]$ respectively.
(b) Show that

$$\int_a^b \left(\int_c^d F(x,y)dy \right) dx = \int_c^d \left(\int_a^b F(x,y)dx \right) dy.$$

Hint: the integrals exist by (a). The identity holds for constant F; derive the general case by subdividing the rectangle into small rectangles and approximating F suitably.

Solution. **(a)** The argument is symmetric for both x and y. It suffices to show one. Note the following bounds,

$$\begin{aligned}\left| \int_a^b F(x,y)\,\mathrm{d}x - \int_a^b F(x,y_0)\,\mathrm{d}x \right| &= \left| \int_a^b (F(x,y) - F(x,y_0))\,\mathrm{d}y \right| \\ &\leq \int_a^b |F(x,y) - F(x,y_0)|\,\mathrm{d}x.\end{aligned}$$

This last thing would be an issue but we know that F is a continuous function over a compact set and thus is uniformly continuous. Thus $\forall \varepsilon > 0$ there exists a $\delta > 0$ such that if two points are within delta of each other $|F(\vec{x}) - F(\vec{y})| < \frac{\varepsilon}{b-a}$. Thus for all $|y - y_0| < \delta$ we have that

$$\left| \int_a^b F(x,y)\,\mathrm{d}x - \int_a^b F(x,y_0)\,\mathrm{d}x \right| < \varepsilon$$

thus we have uniform cont.

(b): You can approximate F uniformly by a peicewise constant function G. Formula holds for each G and then by uniform convergance the integerals should all equal each other.

Problem 4.3

Let $\alpha, \beta : [0,1] \to \mathbb{C}$ be piecewise smooth paths and $G : \alpha^* \times \beta^* \to \mathbb{C}$ continuous. Show that

$$\int_\alpha \left(\int_\beta G(z,w)dw \right) dz = \int_\beta \left(\int_\alpha G(z,w)dz \right) dw.$$

Hint: use Problem 4.2 to show that these integrals exist and are equal.

Solution. Convert the iterated path integrals into integrals over $[0,1]^2 \to \mathbb{C}$.